MALAYSIAN PALM OIL BOARD (MPOB) · PALMNEX

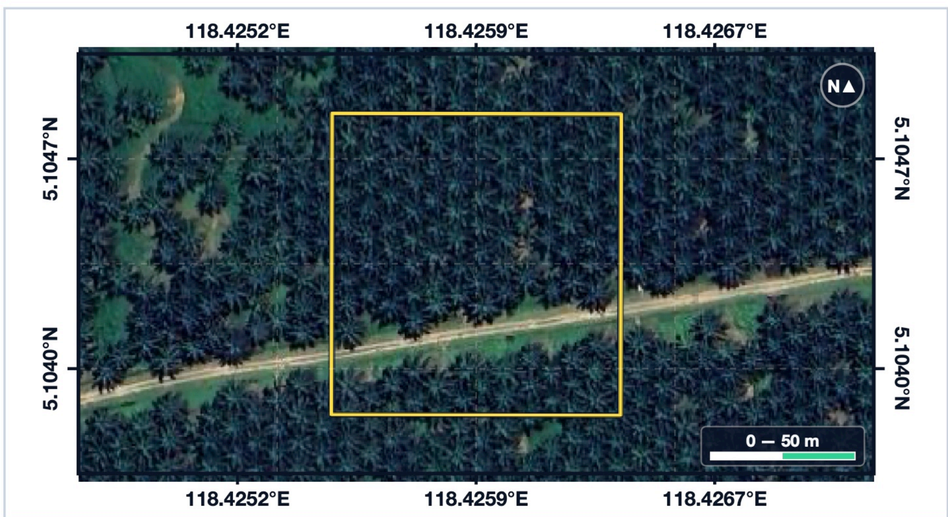
EXECUTIVE AGRONOMIC & SPATIAL FOLIAR DIAGNOSTIC REPORT

COMPREHENSIVE INLAND OPENS PROTOCOL | SITE: LAHAD DATU - BOUNDARY + 36 TRIAL PLOTS | BLOCK 15 (1.00 HA)

1.0 ESTATE & BLOCK GEOINFORMATION

Estate / Plantation Name	Lahad Datu - Boundary + 36 Trial Plots
Selected Block Identifier	Block 15
Geodetic Planted Area	1.00 ha (148 palms)
Centroid Coordinates (WGS84)	5.10437°N, 118.42591°E
Pedological Soil Unit & Series	Inland (Mineral Soil)
Stand Age & Planting Density	12 Years · 148 palms/ha
Satellite Sensor & Resolution	Sentinel-1 SAR & Sentinel-2 MSI (10m)
Agronomic Decision Framework	MPOB Leaf 17 & OPENS Response Matrix

FIG 1.1 — SELECTED BLOCK SPATIAL CONTEXT



2.0 BLOCK FOLIAR MACRO- & MICRO-NUTRIENT DIAGNOSTICS & OPENS ECONOMETRIC PRESCRIPTION

MPOB LEAF 17 THRESHOLDS

TARGET COST/REVENUE RATIO	PREDICTED FFB YIELD	PROJECTED GROSS REVENUE	VALUATION BASIS
CRR = 0.30	22.50 t/ha/yr	RM 337,500.00 / yr	1.00 ha × RM 800/t FFB

Nutrient Element	Block Mean	Spatial Range	Critical Level	Diagnostic Status	Rec. (kg/palm)	Rec. (kg/ha)
Nitrogen (N)	2.36%	2.25–2.57%	2.50%	DEFICIENT (95%)	1.00	143.00
Phosphorus (P)	0.127%	0.119–0.136%	0.150%	DEFICIENT (85%)	0.27	38.61
Potassium (K)	0.82%	0.76–0.95%	1.00%	DEFICIENT (82%)	3.00	429.00
Magnesium (Mg)	0.31%	0.26–0.35%	0.24%	SUFFICIENT (128%)	0.00	0.00
Boron (B)	15.4 ppm	11.8–20.6 ppm	15.0 ppm	SUFFICIENT (103%)	0.10	14.30
Calcium (Ca)	0.71%	0.60–0.87%	0.60%	SUFFICIENT (118%)	0.50	71.50

Recommended Compound Formulation: MPOB F2 Super K (N:7, P:3, K:30, Mg:0, B:0.5)

Cross-Reference: Page 2 (OPENS Matrices) · Page 3 (Block Rasters) · Page 4 (Estate Rasters)

3.0 FOLIAR SUFFICIENCY PROFILE, CANOPY PHYSIOLOGICAL STRESS & AGRONOMIC INTERPRETATION

BLOCK 15

FIG 1.2 — NUTRIENT SUFFICIENCY INDEX (% OF CRITICAL THRESHOLD)

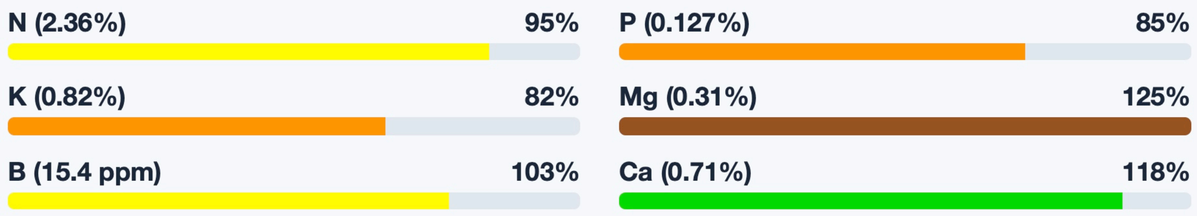
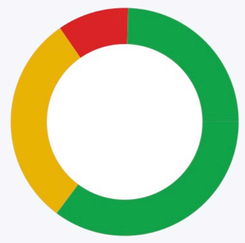


FIG 1.3 — CANOPY STRESS INDEX

- Stressed Canopy: 10.0% (0.1 ha)
- Moderate Vigour: 30.0% (0.3 ha)
- Healthy Canopy: 60.0% (0.6 ha)



AGRONOMIC DIAGNOSTIC SYNTHESIS: Quantitative Sentinel-2 foliar diagnostics for **Block 15** (1.00 ha) indicate sufficient foliar status across Magnesium (Mg), Boron (B), Calcium (Ca), whereas sub-optimal concentrations are detected for **Nitrogen (N), Phosphorus (P), Potassium (K)** relative to MPOB Leaf 17 critical thresholds. Based on the Inland OPENS econometric model (Page 2) at a target Cost/Revenue Ratio (CRR) of **0.30**, an annual prescription of **1.00 kg N/palm** (143.0 kg/ha) and **3.00 kg K/palm** (429.0 kg/ha) is recommended to achieve a predicted FFB yield of **22.50 t/ha/yr**.

4.0 RECOMMENDED SPLIT-APPLICATION SCHEDULE

MPOB BEST MANAGEMENT PRACTICE · BLOCK 15

Application Window	Split Share	N Rate (kg/palm)	K Rate (kg/palm)	Recommended Placement Zone
Round 1 (Mar–Apr · Post-Monsoon)	40%	0.40	1.20	Weed Circle / Frond Pile
Round 2 (Jul–Aug · Mid-Year)	35%	0.35	1.05	Inter-Row Frond Stack
Round 3 (Oct–Nov · Pre-Monsoon)	25%	0.25	0.75	Outer Harvester Path

INLAND SOIL ANALYSIS INPUT (OPENS) & RESPONSE SURFACE MATRICES

STAND & EDAPHIC CHARACTERISTICS, FFB YIELD PREDICTION & GROSS REVENUE GAIN MATRICES | BLOCK 15

TABLE 2.1 — STAND & PHYSICAL SOIL PARAMETERS

Palm Stand Age (Year)	12 yrs
Planting Density (palms/ha)	148 palms/ha
Soil Drainage Class	Class 1
Soil Consistency Class	Class 1
Terrain Slope Factor	0.5
Root Growth Impedance Class	0.2

TABLE 2.2 — EDAPHIC CHEMICAL & CLIMATIC INPUTS

Soil Organic Matter (SOM %)	2 %
Silt Particle Fraction (%)	18 %
Extractable Potassium K (cmol/kg)	0.13 cmol/kg
Total Extractable Bases (TEB)	1.3 cmol/kg
Mean Annual Rainfall (mm/yr)	2000 mm/yr
Pedological Soil Group	Inland Mineral Soil

FFB PRICE / TONNE	SOA COST / TONNE	KCL COST / TONNE	SELECTED N & K RATE	CHOSEN CRR RATIO	PREDICTED YIELD
RM 800	RM 1200	RM 1600	N:1.0 · K:3.0 kg	0.30	22.50 t/ha/yr

TABLE 2.3 — OPENS RESPONSE SURFACE MATRIX: PREDICTED FFB YIELD (tonnes/ha/yr)

Selected Treatment (N=1.0, K=3.0) · Optimal CRR = 0.30 Zone

POTASSIUM (K) FERTILIZER RATE (KG/PALM/YR) →

N \ K (kg)	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
N = 0.0	14.38	14.85	15.23	15.49	15.68	15.80	15.86	15.88	15.86	15.81	15.73
N = 0.5	15.22	15.76	16.22	16.54	16.80	16.98	17.11	17.18	17.21	17.21	17.18
N = 1.0	15.90	16.52	17.05	17.44	17.78	18.01	**18.20**	18.33	18.42	18.47	18.49
N = 1.5	16.32	16.99	17.59	18.04	18.43	18.72	18.96	19.14	19.27	19.36	19.42
N = 2.0	16.63	17.37	18.03	18.54	18.99	19.33	19.63	19.85	20.04	20.17	20.27
N = 2.5	16.81	17.60	18.31	18.87	19.37	19.75	20.09	20.35	20.57	**20.74**	20.88
N = 3.0	16.92	17.76	18.53	19.14	19.68	20.11	20.50	20.80	21.06	21.26	**21.44**
N = 3.5	**16.96**	17.85	18.66	19.31	19.89	20.36	20.78	21.11	21.41	21.64	**21.84**
N = 4.0	16.97	17.90	18.76	19.44	20.07	20.57	21.03	21.39	21.72	21.98	**22.21**
N = 4.5	16.95	17.92	18.81	19.53	20.18	20.71	21.20	21.59	21.94	**22.22**	22.48
N = 5.0	16.91	17.91	18.84	**19.59**	20.28	20.83	21.35	21.76	**22.14**	**22.45**	22.73
N = 5.5	16.86	17.89	18.84	19.62	**20.33**	**20.91**	**21.45**	**21.89**	**22.29**	22.61	22.91
N = 6.0	16.80	17.85	18.84	19.64	20.38	20.98	21.54	22.00	22.42	22.76	23.08

TABLE 2.4 — OPENS RESPONSE SURFACE MATRIX: GROSS REVENUE GAIN (RM/ha/yr)

Selected Treatment (N=1.0, K=3.0) · Optimal CRR = 0.30 Zone

POTASSIUM (K) FERTILIZER RATE (KG/PALM/YR) →

N \ K (kg)	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
N = 0.0	0.00	376.00	680.00	888.00	1,040	1,136	1,184	1,200	1,184	1,144	1,080
N = 0.5	672.00	1,104	1,472	1,728	1,936	2,080	2,184	2,240	2,264	2,264	2,240
N = 1.0	1,216	1,712	2,136	2,448	2,720	2,904	**3,056**	3,160	3,232	3,272	3,288
N = 1.5	1,552	2,088	2,568	2,928	3,240	3,472	3,664	3,808	3,912	3,984	4,032
N = 2.0	1,800	2,392	2,920	3,328	3,688	3,960	4,200	4,376	4,528	4,632	4,712
N = 2.5	1,944	2,576	3,144	3,592	3,992	4,296	4,568	4,776	4,952	**5,088**	5,200
N = 3.0	2,032	2,704	3,320	3,808	4,240	4,584	4,896	5,136	5,344	5,504	**5,648**
N = 3.5	**2,064**	2,776	3,424	3,944	4,408	4,784	5,120	5,384	5,624	5,808	**5,968**
N = 4.0	2,072	2,816	3,504	4,048	4,552	4,952	5,320	5,608	5,872	6,080	**6,264**
N = 4.5	2,056	2,832	3,544	4,120	4,640	5,064	5,456	5,768	6,048	**6,272**	6,480
N = 5.0	2,024	2,824	3,568	**4,168**	4,720	5,160	5,576	5,904	**6,208**	**6,456**	6,680
N = 5.5	1,984	2,808	3,568	4,192	**4,760**	**5,224**	**5,656**	**6,008**	**6,328**	6,584	6,824
N = 6.0	1,936	2,776	3,568	4,208	4,800	5,280	5,728	6,096	6,432	6,704	6,960

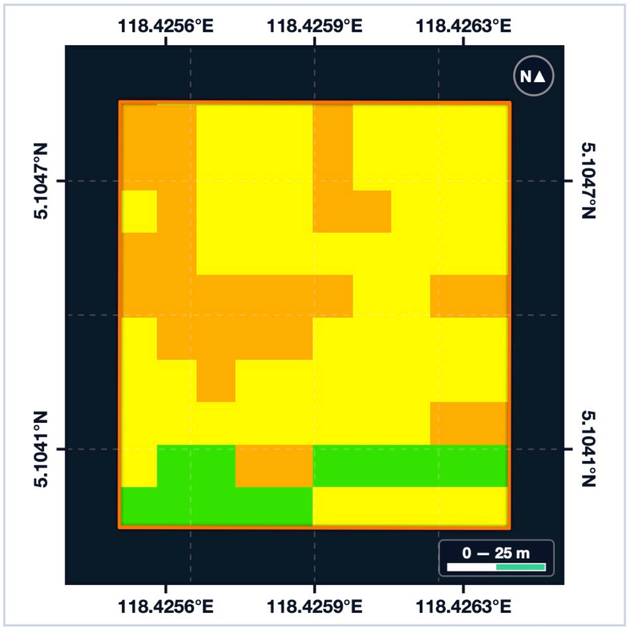
METHODOLOGICAL NOTE (OPENS RESPONSE SURFACE MODELLING): The Oil Palm Efficient Nutrient System (OPENS) response matrices evaluate 143 N × K factorial treatment combinations (N: 0.0–6.0 kg/palm; K: 0.0–5.0 kg/palm) parameterized by site-specific pedological, edaphic, and climatic variables (Tables 2.1–2.2). Shaded iso-zones denote combinations achieving the optimal marginal Cost/Revenue Ratio (CRR ≈ 0.30), while the highlighted blue cell designates the selected operational prescription for Block 15.

SELECTED BLOCK SPATIAL FOLIAR NUTRIENT ANALYSIS (N, P, K, MG, CA, B)

BLOCK-LEVEL FOLIAR DISTRIBUTION & MPOB LEAF 17 CRITICAL THRESHOLDS | CRS: WGS 84 (EPSG:4326) | RESOLUTION: 10m SENTINEL-2 MSI

NITROGEN (%) SPATIAL ANALYSIS

DEFICIENT



CLASSIFICATION LEGEND	AREA %
≤ 2.10 %	Deficient (0%)
2.10–2.30 %	Low (28%)
2.30–2.50 %	Marginal (60%)
2.50–2.70 %	Optimum (12%)
2.70–2.90 %	High (0%)
> 2.90 %	Excessive (0%)

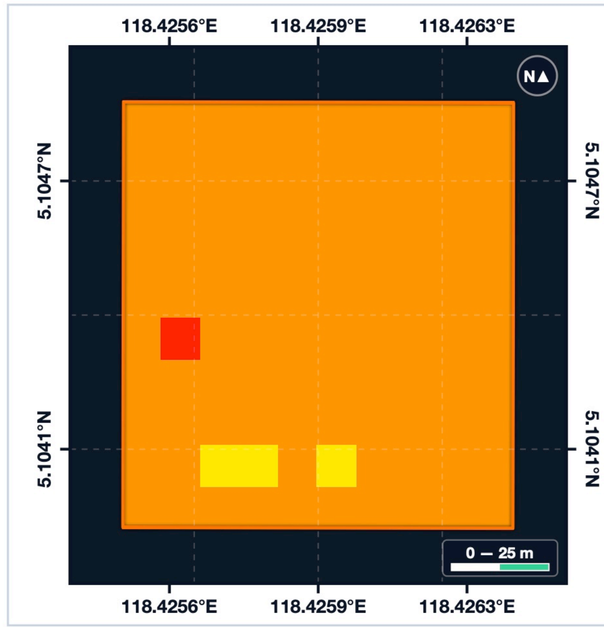
STATISTICAL SUMMARY (BLOCK)

Mean ± SD:	2.36 ± 0.08 %
Min–Max Range:	2.25–2.57 %
Critical Level:	2.50%
Adequate (≥Crit):	12.0% (CV 3.5%)

Spatial foliar diagnostics across **Block 15** reveal a sub-optimal mean nitrogen concentration of **2.36 %** (range 2.25–2.57 %, CV 3.5%), below the MPOB critical threshold (2.50%). Localized deficiency affects **88.0%** of the canopy zone, warranting targeted corrective supplementation.

PHOSPHORUS (%) SPATIAL ANALYSIS

DEFICIENT



CLASSIFICATION LEGEND	AREA %
≤ 0.120 %	Deficient (1%)
0.120–0.135 %	Low (98%)
0.135–0.150 %	Marginal (1%)
0.150–0.165 %	Optimum (0%)
0.165–0.180 %	High (0%)
> 0.180 %	Excessive (0%)

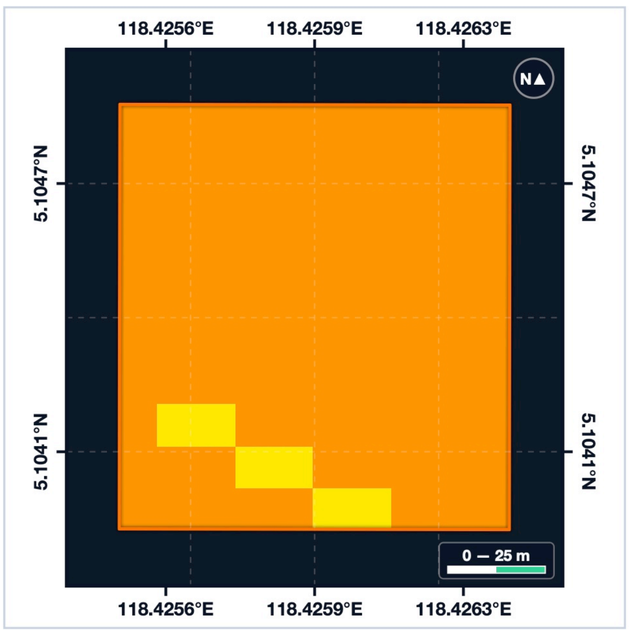
STATISTICAL SUMMARY (BLOCK)

Mean ± SD:	0.127 ± 0.005 %
Min–Max Range:	0.119–0.136 %
Critical Level:	0.150%
Adequate (≥Crit):	0.0% (CV 3.5%)

Spatial foliar diagnostics across **Block 15** reveal a sub-optimal mean phosphorus concentration of **0.127 %** (range 0.119–0.136 %, CV 3.5%), below the MPOB critical threshold (0.150%). Localized deficiency affects **100.0%** of the canopy zone, warranting targeted corrective supplementation.

POTASSIUM (%) SPATIAL ANALYSIS

DEFICIENT



CLASSIFICATION LEGEND	AREA %
≤ 0.70 %	Deficient (0%)
0.70–0.85 %	Low (94%)
0.85–1.00 %	Marginal (6%)
1.00–1.15 %	Optimum (0%)
1.15–1.30 %	High (0%)
> 1.30 %	Excessive (0%)

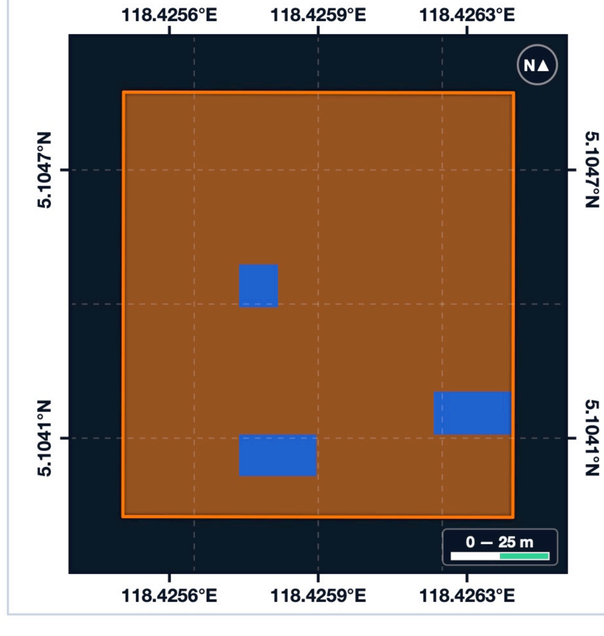
STATISTICAL SUMMARY (BLOCK)

Mean ± SD:	0.82 ± 0.02 %
Min–Max Range:	0.76–0.95 %
Critical Level:	1.00%
Adequate (≥Crit):	0.0% (CV 3.0%)

Spatial foliar diagnostics across **Block 15** reveal a sub-optimal mean potassium concentration of **0.82 %** (range 0.76–0.95 %, CV 3.0%), below the MPOB critical threshold (1.00%). Localized deficiency affects **100.0%** of the canopy zone, warranting targeted corrective supplementation.

MAGNESIUM (%) SPATIAL ANALYSIS

SUFFICIENT



CLASSIFICATION LEGEND	AREA %
≤ 0.20 %	Deficient (0%)
0.20–0.22 %	Low (0%)
0.22–0.24 %	Marginal (0%)
0.24–0.26 %	Optimum (0%)
0.26–0.28 %	High (5%)
> 0.28 %	Excessive (95%)

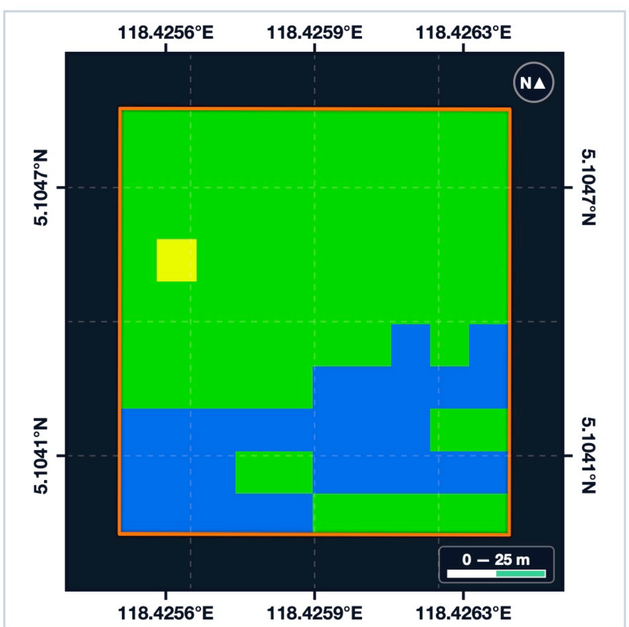
STATISTICAL SUMMARY (BLOCK)

Mean ± SD:	0.31 ± 0.02 %
Min–Max Range:	0.26–0.35 %
Critical Level:	0.24%
Adequate (≥Crit):	100.0% (CV 6.7%)

Spatial foliar mapping across **Block 15** indicates an average magnesium level of **0.31 %** (range 0.26–0.35 %, CV 6.7%), exceeding the MPOB Leaf 17 critical threshold (0.24%). Approximately **100.0%** of the analyzed canopy area exhibits optimal-to-high sufficiency, requiring standard maintenance fertilization.

CALCIUM (%) SPATIAL ANALYSIS

SUFFICIENT



CLASSIFICATION LEGEND	AREA %
≤ 0.40 %	Deficient (0%)
0.40–0.50 %	Low (0%)
0.50–0.60 %	Marginal (1%)
0.60–0.75 %	Optimum (71%)
0.75–0.90 %	High (28%)
> 0.90 %	Excessive (0%)

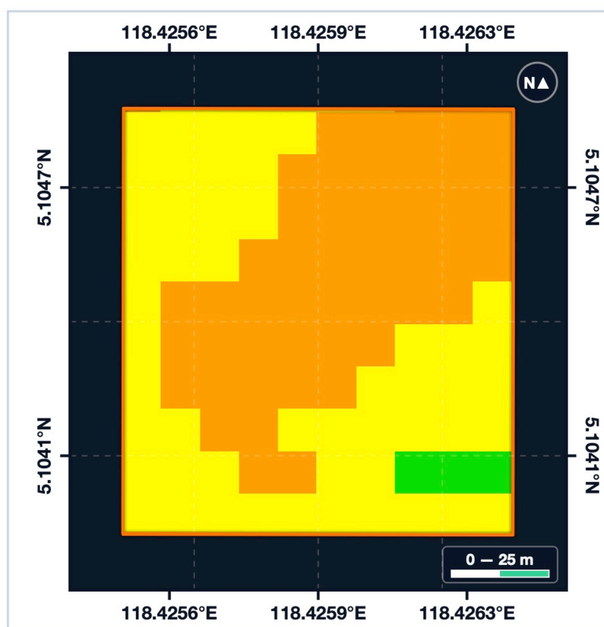
STATISTICAL SUMMARY (BLOCK)

Mean ± SD:	0.71 ± 0.07 %
Min–Max Range:	0.60–0.87 %
Critical Level:	0.60%
Adequate (≥Crit):	99.0% (CV 10.2%)

Spatial foliar mapping across **Block 15** indicates an average calcium level of **0.71 %** (range 0.60–0.87 %, CV 10.2%), exceeding the MPOB Leaf 17 critical threshold (0.60%). Approximately **99.0%** of the analyzed canopy area exhibits optimal-to-high sufficiency, requiring standard maintenance fertilization.

BORON (PPM) SPATIAL ANALYSIS

SUFFICIENT



CLASSIFICATION LEGEND	AREA %
≤ 10.0 ppm	Deficient (0%)
10.0–15.0 ppm	Low (47%)
15.0–20.0 ppm	Marginal (50%)
20.0–30.0 ppm	Optimum (3%)
30.0–40.0 ppm	High (0%)
> 40.0 ppm	Excessive (0%)

STATISTICAL SUMMARY (BLOCK)

Mean ± SD:	15.4 ± 2.1 ppm
Min–Max Range:	11.8–20.6 ppm
Critical Level:	15.0 ppm
Adequate (≥Crit):	54.0% (CV 13.8%)

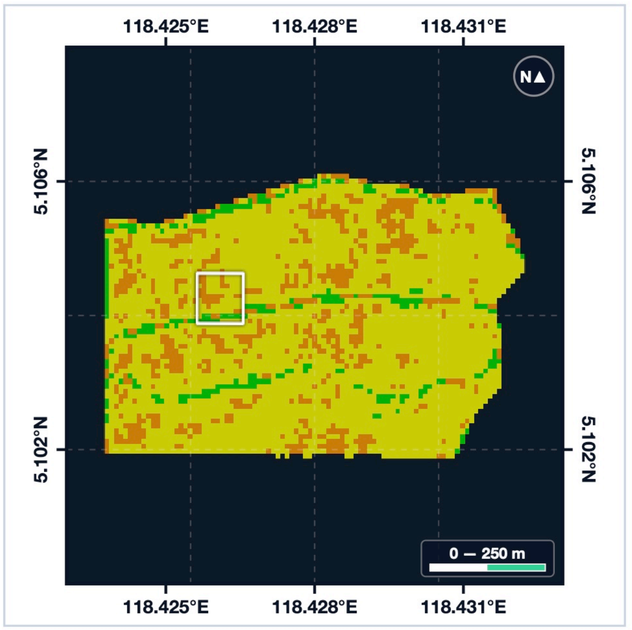
Spatial foliar mapping across **Block 15** indicates an average boron level of **15.4 ppm** (range 11.8–20.6 ppm, CV 13.8%), exceeding the MPOB Leaf 17 critical threshold (15.0 ppm). Approximately **54.0%** of the analyzed canopy area exhibits optimal-to-high sufficiency, requiring standard maintenance fertilization.

WHOLE ESTATE SPATIAL FOLIAR NUTRIENT ANALYSIS (N, P, K, MG, CA, B)

ESTATE-WIDE FOLIAR DISTRIBUTION & MPOB LEAF 17 CRITICAL THRESHOLDS | CRS: WGS 84 (EPSG:4326) | RESOLUTION: 10m SENTINEL-2 MSI

NITROGEN (%) SPATIAL ANALYSIS

DEFICIENT



CLASSIFICATION LEGEND	AREA %
≤ 2.10 %	Deficient (0%)
2.10–2.30 %	Low (18%)
2.30–2.50 %	Marginal (76%)
2.50–2.70 %	Optimum (6%)
2.70–2.90 %	High (0%)
> 2.90 %	Excessive (0%)

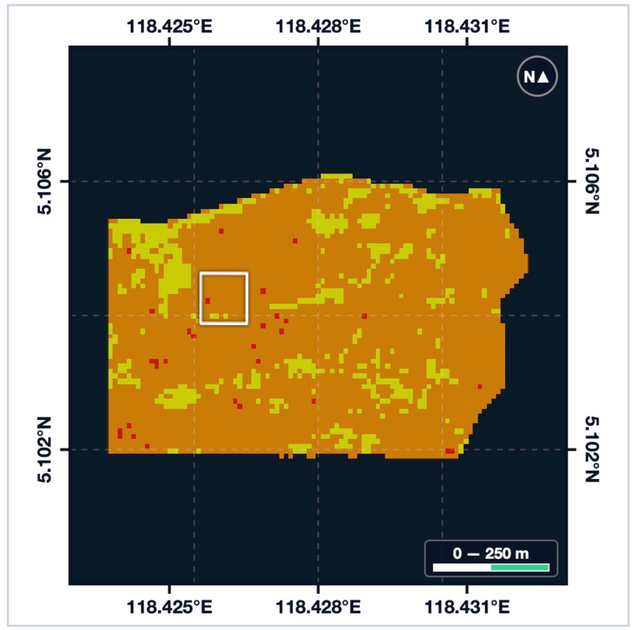
STATISTICAL SUMMARY (ESTATE)

Mean ± SD:	2.37 ± 0.07 %
Min–Max Range:	2.15–2.59 %
Critical Level:	2.50%
Adequate (≥Crit):	6.2% (CV 3.0%)

Spatial foliar diagnostics across **Lahad Datu - Boundary + 36 Trial Plots Estate** reveal a sub-optimal mean nitrogen concentration of **2.37 %** (range 2.15–2.59 %, CV 3.0%), below the MPOB critical threshold (2.50%). Localized deficiency affects **93.8%** of the canopy zone, warranting targeted corrective supplementation.

PHOSPHORUS (%) SPATIAL ANALYSIS

DEFICIENT



CLASSIFICATION LEGEND	AREA %
≤ 0.120 %	Deficient (1%)
0.120–0.135 %	Low (89%)
0.135–0.150 %	Marginal (10%)
0.150–0.165 %	Optimum (0%)
0.165–0.180 %	High (0%)
> 0.180 %	Excessive (0%)

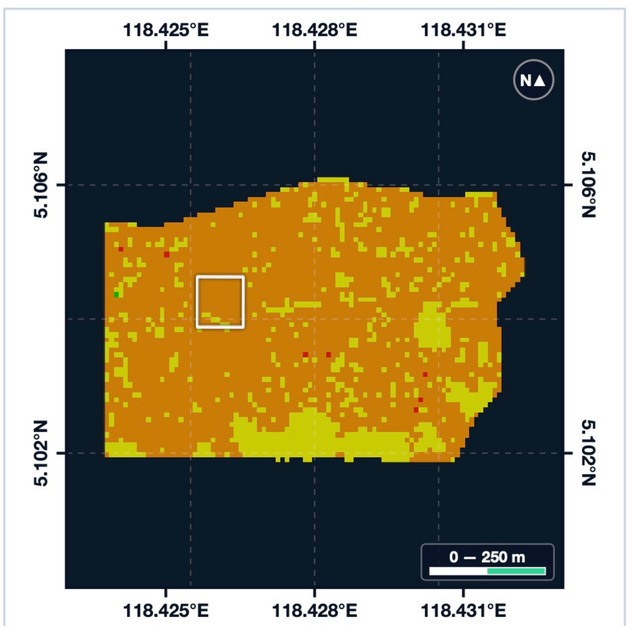
STATISTICAL SUMMARY (ESTATE)

Mean ± SD:	0.129 ± 0.005 %
Min–Max Range:	0.118–0.146 %
Critical Level:	0.150%
Adequate (≥Crit):	0.0% (CV 4.0%)

Spatial foliar diagnostics across **Lahad Datu - Boundary + 36 Trial Plots Estate** reveal a sub-optimal mean phosphorus concentration of **0.129 %** (range 0.118–0.146 %, CV 4.0%), below the MPOB critical threshold (0.150%). Localized deficiency affects **100.0%** of the canopy zone, warranting targeted corrective supplementation.

POTASSIUM (%) SPATIAL ANALYSIS

DEFICIENT



CLASSIFICATION LEGEND	AREA %
≤ 0.70 %	Deficient (0%)
0.70–0.85 %	Low (80%)
0.85–1.00 %	Marginal (20%)
1.00–1.15 %	Optimum (0%)
1.15–1.30 %	High (0%)
> 1.30 %	Excessive (0%)

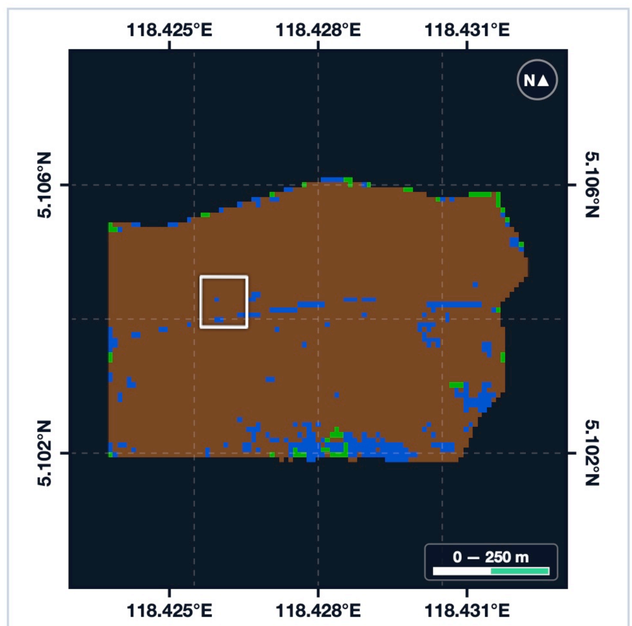
STATISTICAL SUMMARY (ESTATE)

Mean ± SD:	0.83 ± 0.04 %
Min–Max Range:	0.64–1.00 %
Critical Level:	1.00%
Adequate (≥Crit):	0.0% (CV 5.2%)

Spatial foliar diagnostics across **Lahad Datu - Boundary + 36 Trial Plots Estate** reveal a sub-optimal mean potassium concentration of **0.83 %** (range 0.64–1.00 %, CV 5.2%), below the MPOB critical threshold (1.00%). Localized deficiency affects **100.0%** of the canopy zone, warranting targeted corrective supplementation.

MAGNESIUM (%) SPATIAL ANALYSIS

SUFFICIENT



CLASSIFICATION LEGEND	AREA %
≤ 0.20 %	Deficient (0%)
0.20–0.22 %	Low (0%)
0.22–0.24 %	Marginal (0%)
0.24–0.26 %	Optimum (1%)
0.26–0.28 %	High (7%)
> 0.28 %	Excessive (92%)

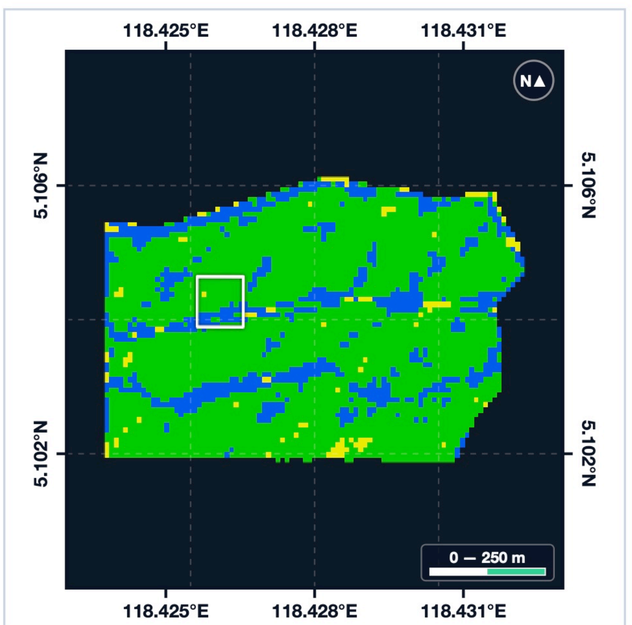
STATISTICAL SUMMARY (ESTATE)

Mean ± SD:	0.30 ± 0.02 %
Min–Max Range:	0.24–0.37 %
Critical Level:	0.24%
Adequate (≥Crit):	100.0% (CV 6.4%)

Spatial foliar mapping across **Lahad Datu - Boundary + 36 Trial Plots Estate** indicates an average magnesium level of **0.30 %** (range 0.24–0.37 %, CV 6.4%), exceeding the MPOB Leaf 17 critical threshold (0.24%). Approximately **100.0%** of the analyzed canopy area exhibits optimal-to-high sufficiency, requiring standard maintenance fertilization.

CALCIUM (%) SPATIAL ANALYSIS

SUFFICIENT



CLASSIFICATION LEGEND	AREA %
≤ 0.40 %	Deficient (0%)
0.40–0.50 %	Low (0%)
0.50–0.60 %	Marginal (3%)
0.60–0.75 %	Optimum (79%)
0.75–0.90 %	High (18%)
> 0.90 %	Excessive (0%)

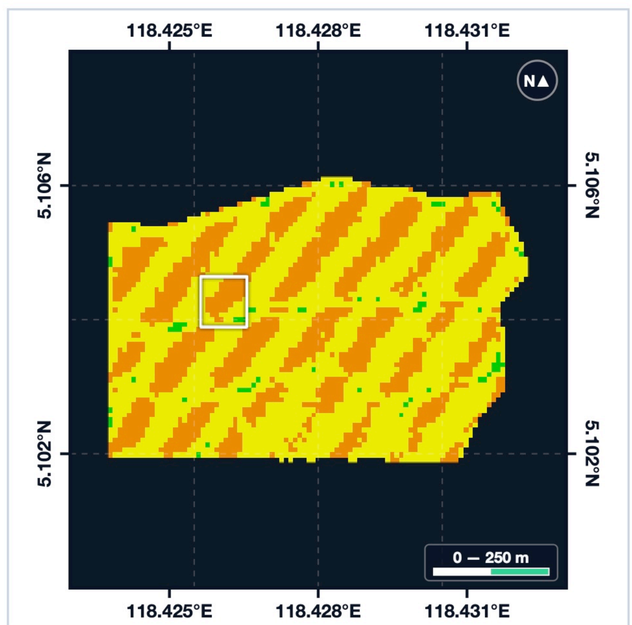
STATISTICAL SUMMARY (ESTATE)

Mean ± SD:	0.70 ± 0.06 %
Min–Max Range:	0.54–0.89 %
Critical Level:	0.60%
Adequate (≥Crit):	97.4% (CV 8.3%)

Spatial foliar mapping across **Lahad Datu - Boundary + 36 Trial Plots Estate** indicates an average calcium level of **0.70 %** (range 0.54–0.89 %, CV 8.3%), exceeding the MPOB Leaf 17 critical threshold (0.60%). Approximately **97.4%** of the analyzed canopy area exhibits optimal-to-high sufficiency, requiring standard maintenance fertilization.

BORON (PPM) SPATIAL ANALYSIS

SUFFICIENT



CLASSIFICATION LEGEND	AREA %
≤ 10.0 ppm	Deficient (0%)
10.0–15.0 ppm	Low (36%)
15.0–20.0 ppm	Marginal (63%)
20.0–30.0 ppm	Optimum (1%)
30.0–40.0 ppm	High (0%)
> 40.0 ppm	Excessive (0%)

STATISTICAL SUMMARY (ESTATE)

Mean ± SD:	15.8 ± 2.0 ppm
Min–Max Range:	10.1–21.5 ppm
Critical Level:	15.0 ppm
Adequate (≥Crit):	64.5% (CV 12.4%)

Spatial foliar mapping across **Lahad Datu - Boundary + 36 Trial Plots Estate** indicates an average boron level of **15.8 ppm** (range 10.1–21.5 ppm, CV 12.4%), exceeding the MPOB Leaf 17 critical threshold (15.0 ppm). Approximately **64.5%** of the analyzed canopy area exhibits optimal-to-high sufficiency, requiring standard maintenance fertilization.